Percentage Increases & Decreases

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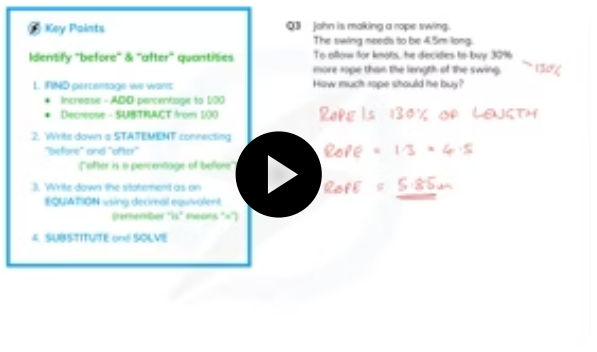

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Percentage Increases & Decreases

How do I increase by a percentage?

- A percentage **increase** makes an amount **bigger** by **adding** that percentage on to itself
- **Without a calculator**, use the **basic percentages** methods to find the percentage you are increasing by
  - Then **add** this on to the original amount
  - To increase 30 by 10%
    - 10% of 30 is 3
    - 30 + 3 = 33
    - This is equivalent to finding 110% of 30
- **With a calculator** it is more efficient to use **multipliers**
  - A multiplier is the **decimal equivalent** of a percentage
    - A **percentage** can be converted **to a decimal** by **dividing by 100**
  - When **increasing** by a percentage, we are finding a percentage **greater than 100%**
  - To increase 80 by 15%
    - We are finding **115%** of 80, so the multiplier is **1.15**
    - 1.15 × 80 = 92

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

Help me understand

- A percentage **decrease** makes an amount **smaller** by **subtracting** that percentage from itself
- **Without a calculator**, use the methods outlined in **Basic Percentages** to find the percentage you are decreasing by
  - Then **subtract** this from the original amount
  - To decrease 30 by 10%
    - 10% of 30 is 3
    - 30 - 3 = 27
    - This is equivalent to finding 90% of 30
      - Because 100% - 10% = 90%

- **With a calculator** it is more efficient to use **multipliers**
  - When **decreasing** by a percentage, we are finding a percentage **smaller than 100%**
  - To decrease 80 by 15%
    - We are finding **85%** of 80, so the multiplier is **0.85**
      - Because 100% - 15% = 85%
    - 0.85 × 80 = 68

Worked Example

(a) Increase 200 kg by 21%.

Answer:

Method 1 (non-calculator)

By first finding 10% and 1%, find 21% of 200

10% of 200 = 20
1% of 200 = 2
21% of 200 = 20 + 20 + 2 = 42

Add this to the original amount

200 + 42

242 kg

Method 2 (calculator)

An increase by 21% is equivalent to finding 121% of the original amount
So the multiplier is 1.21

1.21 × 200

242 kg

(b) An item that costs $ 500 is discounted by 35%.

Find the new price of the item.

Answer:

A discount of 35% means the price decreases by 35%

Method 1 (non-calculator)

5% of 500 = 25

35% of 500 = 50 + 50 + 50 + 25 = 175

Subtract this from the original amount

500 - 175

$ 325

Method 2 (calculator)

A decrease of 35% is equivalent to finding 65% of the original amount (100 - 35 = 65)
So the multiplier is 0.65

500 × 0.65

$ 325

How do I deal with repeated percentage changes?

- In some problems there may be **several** changes by a percentage
- For example,
  - A shop **increases** the price of a product costing £80 **by 10%**,
    - equivalent to a multiplier of × 1.10
  - and then **discounts** the product **by 15%**,
    - equivalent to a multiplier of × 0.85
  - and then **discounts** the product by a **further 20%**
    - equivalent to a multiplier of × 0.80
- You can either:
  - **Multiply** the starting amount **by each multiplier in turn**
    - (((80 × 1.10) × 0.85) × 0.80) = £59.84
  - Or **combine** the multipliers first and then multiply by the "combined multiplier"
    - 1.10 × 0.85 × 0.80 = 0.748
      - This shows it is equivalent to 74.8% of the original amount, or a discount of 25.2%
    - 80 × 0.748 = £59.84
- In general, for n multipliers of values $m_1, m_2, \ldots, m_n$
  - The combined multiplier is $m_1 \times m_2 \times \ldots \times m_n$

How do I find a percentage change?

- The **multiplier** that was used for a **percentage change** can be found using the formula:
  - $m = \dfrac{\text{Amount after}}{\text{Amount before}}$
- The value of m corresponds to the **multiplier** for the percentage change
  - A value **greater than 1** is a percentage **increase**
    - **1.05** corresponds to an **increase by 5%**

- Percentage Change = $\frac{\text{After} - \text{Before}}{\text{Before}} \times 100$
- A **positive** value is a percentage **increase**
  - An answer of 12 means an increase of 12%
- A **negative** value is a percentage **decrease**
  - An answer of -28 means a decrease of 28%

How do I find a percentage profit or loss?

- Similar strategies to the above can be used to find the **percentage profit or loss**
- Shops buy or produce items at a "cost price" and sell them at a "selling price"
- Using a **multiplier** method:
  - $m = \frac{\text{Selling Price}}{\text{Cost Price}}$
  - A value **greater than 1** is a **profit**
    - **1.05** corresponds to a **5% profit**
  - A value **less than 1** is a **loss**
    - **0.75** corresponds to a **25% loss**
- **Alternatively** you can use the formula:
  - Percentage Profit = $\frac{\text{Selling Price} - \text{Cost Price}}{\text{Cost Price}} \times 100$
  - A **positive** value is a **profit**
    - An answer of 12 means a 12% profit
  - A **negative** value is a **loss**
    - An answer of -28 means a 28% loss

Examiner Tips and Tricks

- Use "common sense" to check your answer!
  - If an item is sold for more than it was bought for, you are expecting a profit, not a loss

Worked Example

The number of students in a school changes from 250 to 310.

Describe the percentage change in number of students.

Answer:

Method 1

Use the formula $m = \frac{\text{Amount after}}{\text{Amount before}}$

$$\frac{310}{250} = 1.24$$

A percentage increase of 24%

Method 2

Use the formula Percentage Change = $\frac{\text{After} - \text{Before}}{\text{Before}} \times 100$

$$\frac{310 - 250}{250} \times 100 = 24$$

The value is positive, so this is a percentage increase

A percentage increase of 24%

Worked Example

Sophie purchases a car for $8000 and sells it several years later for $5600.

Describe the percentage profit or loss on the car.

Answer:

Method 1

Use the formula $m = \frac{\text{Selling Price}}{\text{Cost Price}}$

$$\frac{5600}{8000} = 0.7$$

The value is less than 1 so means it is a percentage loss

The selling price was 70% of the cost price, so a loss of 30%

A loss of 30%

Method 2

Use the formula Percentage Profit = $\frac{\text{Selling Price} - \text{Cost Price}}{\text{Cost Price}} \times 100$

$$\frac{5600 - 8000}{8000} \times 100 = -30$$

The value is negative, so this is a percentage loss

A loss of 30%

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